

Student Number: _____ KM AM



KAMBALA

Mathematics Extension 1

HSC Assessment Task 2

Half-Yearly Examination

March 2008

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Answer all questions in the writing booklets provided. **Start each question in a new booklet.**
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- More marks will be awarded to questions involving higher order thinking.

Total marks – 84

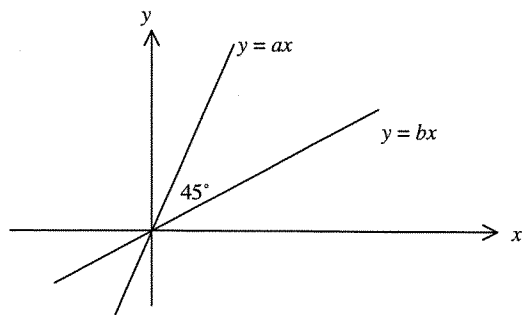
- Attempt Questions 1-7.
- All questions are of equal value.

Question 1 **12 marks** **(Begin a new booklet)** **Marks**

- | | | |
|-----|---|----------|
| (a) | Factorise $64a^3 + \frac{1}{27}b^3$ | 1 |
| (b) | Solve for x : $3^{2x+1} - 28(3^x) + 9 = 0$ | 3 |
| (c) | Evaluate $\int_1^4 x\sqrt{5-x} \, dx$ using the substitution $u = 5 - x$. | 3 |
| (d) | Solve $\frac{4}{5-x} \geq 1$ | 3 |
| (e) | If $P(x) = x^3 - 3kx + 3$ is divisible by $(x - 3)$, find the value of k . | 2 |

Question 2 **12 marks** **(Begin a new booklet)** **Marks**

- (a) In the diagram below, the angle between the lines $y = ax$ and $y = bx$ is 45° . **2**



Show that $b = \frac{a-1}{a+1}$.

- (b) Find $\sum_{a=1}^n \frac{n-a}{n}$ **3**

- (c) The points A(1, -3), B(10, 9) and P(4, 1) are collinear. In what ratio does the point P divide the interval joining A and B? **3**

- (d) Consider the circle $x^2 + y^2 - 2x - 14y + 25 = 0$.

- (i) Show that if the line $y = mx$ intersects the circle in two distinct points, then **2**

$$(1+7m)^2 - 25(1+m^2) > 0$$

- (ii) For what value(s) of m is the line $y = mx$ a tangent to the curve? **2**

Question 3 **12 marks** **(Begin a new booklet)** **Marks**

- (a) Find $\int_{-1}^2 |1-2x| dx$ **2**

- (b) Given that a root of $x + \ln x = 2$ lies close to $x = 1.5$, use Newton's Method to find an approximation for the root. Answer to two decimal places. **3**

- (c) (i) Factorise $3x^3 + 3x^2 - x - 1$. **2**

- (ii) Hence or otherwise, solve the equation $3\tan^3 \theta + 3\tan^2 \theta - \tan \theta - 1 = 0$ for $0^\circ \leq \theta \leq 180^\circ$. **2**

- (d) If $x^4 + 4y^4 = (x^2 + 2y^2 + axy)(x^2 + 2y^2 - axy)$ for all values of x and y , find the value of a . **3**

Question 4 12 marks (Begin a new booklet) Marks

- (a) Find the equation of the tangent to the curve $y = 2^x$ at the point $(1, 2)$. 2

- (b) For $x > 0$, the area bounded by the curve $y = \frac{1}{x}$, the x -axis and the lines $x = e$ and $x = k$ is 1 square unit. 2

Find the value of k if $k < e$.

- (c) Two geometric series are given by:

(A) $1 + (\sqrt{3} - 1) + (\sqrt{3} - 1)^2 + \dots$

(B) $1 + (\sqrt{3} + 1) + (\sqrt{3} + 1)^2 + \dots$

A new series is formed by taking the product of corresponding terms:

ie $1 \times 1 + (\sqrt{3} - 1)(\sqrt{3} + 1) + (\sqrt{3} - 1)^2(\sqrt{3} + 1)^2 + \dots$

- (i) Explain whether this new series is arithmetic, geometric or neither. 2
- (ii) Find the n^{th} term of this series. 2
- (d) By Mathematical Induction, prove that for every positive integer n , $13 \times 6^n + 2$ is divisible by 5. 4

Question 5 12 marks (Begin a new booklet) Marks

- (a) Given α , β and γ are the roots of the equation $2x^3 - 5x^2 + 7x - 1 = 0$, find the values of:

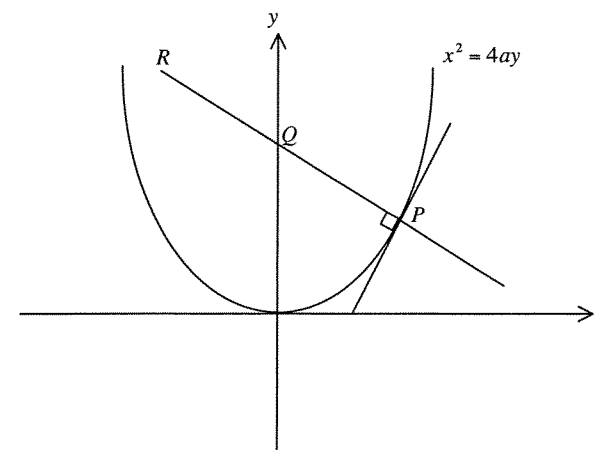
(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ 1

(iii) $(\alpha - 1)(\beta - 1)(\gamma - 1)$ 2

- (b) If $\cos \alpha = \frac{3}{4}$, $0 < \alpha < 90^\circ$ and $\cos \beta = \frac{2}{3}$, $270^\circ < \beta < 360^\circ$, write down the exact value of $\sin \beta$ and hence find the exact value of $\sin(\alpha - \beta)$. 3

- (c)



The normal at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the y -axis at Q and is produced to a point R such that $PQ = QR$.

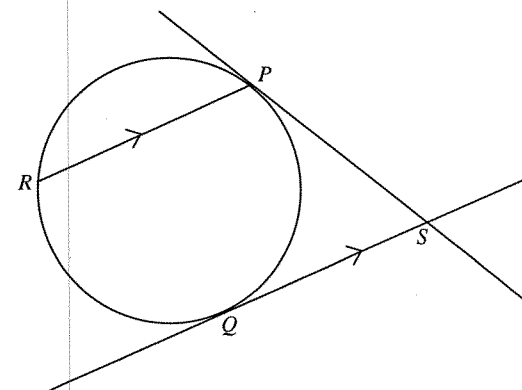
- (i) Given that the equation of the normal at P is $x + py = 2ap + ap^3$, find the co-ordinates of Q . 1
- (ii) Show that R has co-ordinates $(-2ap, ap^2 + 4a)$. 2
- (iii) Show that the locus of R is a parabola and state its vertex. 2

Question 6 12 marks (Begin a new booklet) **Marks**

- (a) Differentiate $\frac{4x+1}{2x-3}$. Hence evaluate $\int_0^1 \frac{dx}{(2x-3)^2}$. 3
- (b) Consider the function $f(x) = \frac{x}{x+2}$.
- (i) Show that $f'(x) > 0$ for all x in the domain. 2
- (ii) State the equation of the horizontal asymptote of $y = f(x)$. 1
- (iii) Without using any further calculus, sketch the graph of $y = f(x)$. 2
- (iv) Explain why $y = f(x)$ has an inverse function $f^{-1}(x)$. 1
- (v) Find an expression for $f^{-1}(x)$. 2
- (vi) Write down the domain of $f^{-1}(x)$. 1

Question 7 12 marks (Begin a new booklet) **Marks**

- (a) (i) Express $\sin A$ and $\cos A$ in terms of t , where $t = \tan \frac{A}{2}$. 1
- (ii) Hence, prove that $\frac{\sin 2A}{1 + \cos 2A} = \tan A$. 3
- (b)



P and Q are points on a circle and the tangents to the circle at P and Q meet at S . R is a point on the circle so that the chord PR is parallel to QS .

Copy the diagram into your answer booklet.

Prove $QP = QR$.

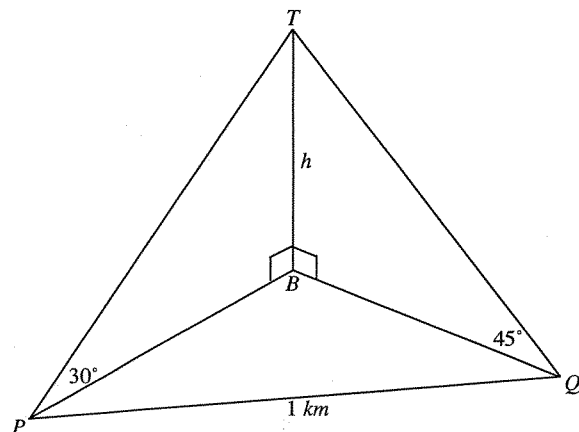
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Question 7 continues next page

Question 7 continued

Marks

(c)



The angle of elevation from a boat at P to a point T at the top of a vertical cliff is 30° . The boat sails 1 km to a second point Q , from which the angle of elevation to T is 45° . Let B be the point at the base of the cliff directly below T and let $h = BT$ be the height of the cliff in metres.

The bearings of B from P and Q are 050°T and 310°T respectively.

- (i) Show that $\angle PBQ = 100^\circ$. 1
- (ii) Find expressions for PB and QB in terms of h . 1
- (iii) Hence show that: 2

$$h^2 = \frac{1000^2}{\cot^2 30 + \cot^2 45 - 2 \cot 30 \cot 45 \cos 100}$$

- (iv) Calculate the height of the cliff, correct to the nearest metre. 1

END OF EXAMINATION

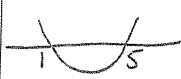
Spare Questions

- (b) Factorise $250 - 2x^3$. 2
- (c) The line $y = \frac{4\sqrt{3}}{3}$ meets the curve $y = \frac{1}{\sqrt{x}} + \sqrt{x}$ at A and B .
- (i) Show that the x co-ordinates of A and B are $\frac{1}{3}$ and 3 respectively. 2
- (ii) Show that the area of the region bounded by the line and the curve is $\frac{8\sqrt{3}}{27} u^2$. 2
- (g) Consider the curve $y = \ln x(x+2)$.
- (i) State its domain. 2
- (ii) Find its x -intercepts. 2
- (iii) Hence sketch the curve. 1

Consider the graph of $y = \frac{x^2}{1-x^2}$.

- a) Write down the vertical asymptotes of the function.
- b) Find any turning points of the function and determine their nature.
- c) Show that the function is even.
- d) Describe how the function behaves for large x .
- e) Sketch the graph.
- (d) Find $\int \frac{e^x}{1+e^x} dx$. 1

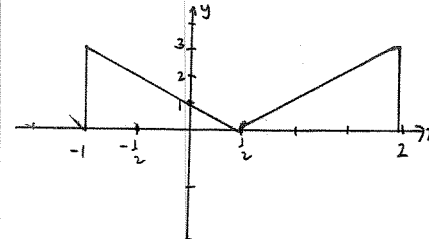
Question	Solutions	Marks	Marking Criteria
1a)	$64a^3 + \frac{1}{27}b^3$ $= (4a)^3 + \left(\frac{1}{3}b\right)^3$ $= \left(4a + \frac{1}{3}b\right) \left[(4a)^2 - (4a)\left(\frac{1}{3}b\right) + \left(\frac{1}{3}b\right)^2\right]$ $= \left(4a + \frac{1}{3}b\right) \left(16a^2 - \frac{4ab}{3} + \frac{b^2}{9}\right)$	1	
1b)	$3^{2x+1} - 28(3^x) + 9 = 0$ $(3^x)^2 \cdot 3 - 28(3^x) + 9 = 0$ let $u = 3^x$ $\therefore 3u^2 - 28u + 9 = 0$ $\therefore (3u - 1)(u - 9) = 0$ $\therefore u = \frac{1}{3}, 9$ $\therefore 3^x = \frac{1}{3} \text{ or } 3^x = 9$ $\therefore 3^x = 3^{-1} \quad 3^x = 3^2$ $\therefore x = -1 \quad \therefore x = 2$ $\therefore x = -1, 2$	1	
1c)	$\int_1^4 x\sqrt{5-x} \, dx$ let $u = 5-x \quad \therefore x = 5-u$ $\therefore \frac{du}{dx} = -1 \Rightarrow dx = \frac{du}{-1} = -du$ $\int_1^4 x\sqrt{5-x} \, dx \quad \begin{matrix} \text{When } x=1, u=4 \\ \text{When } x=4, u=1 \end{matrix}$ $= \int_4^1 (5-u)u^{\frac{1}{2}} \cdot -du$ $= \int_1^4 (5-u)u^{\frac{1}{2}} \, du$ $= \int_1^4 \left(5u^{\frac{1}{2}} - u^{\frac{3}{2}}\right) \, du$	1	

Question	Solutions	Marks	Marking Criteria
1c) ctd	$= \left[\frac{2.5u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{2u^{\frac{5}{2}}}{\frac{5}{2}} \right]_1^4$ $= \left[\frac{10\sqrt{u^3}}{3} - \frac{2\sqrt{u^5}}{5} \right]_1^4$ $= \left\{ \frac{10\sqrt{64}}{3} - \frac{2\sqrt{1024}}{5} \right\} - \left\{ \frac{10}{3} - \frac{2}{5} \right\}$ $= \left\{ \frac{80}{3} - \frac{64}{5} \right\} - \left\{ \frac{10}{3} - \frac{2}{5} \right\}$ $= \frac{70}{3} - \frac{62}{5}$ $= 23\frac{1}{3} - 12\frac{2}{5}$ $= 10\frac{14}{15}$	1	
1d)	$\frac{4}{5-x} \geq 1$ $4(5-x) \geq (5-x)^2, \quad x \neq 5$ $20 - 4x \geq 25 - 10x + x^2$ $0 \geq x^2 - 6x + 5$ $x^2 - 6x + 5 \leq 0$ $(x-1)(x-5) \leq 0$  $\therefore$ by inspection, $1 \leq x \leq 5$ but $x \neq 5$ $\therefore 1 \leq x < 5$	1	

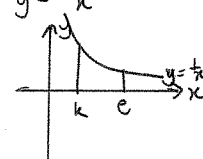
Question	Solutions	Marks	Marking Criteria
1e)	$P(x) = x^3 - 3kx + 3$ divisible by $(x-3)$ $\therefore P(3) = 0$ $\therefore (3)^3 - 3k(3) + 3 = 0$ $27 - 9k + 3 = 0$ $30 - 9k = 0$ $k = \frac{30}{9}$ $k = \frac{10}{3}$	1 1	
2a)	$y = ax \text{ has } m_1 = a$ $y = bx \text{ has } m_2 = b$ $\tan 45^\circ = 1 \quad \text{RTP: } b = \frac{a-1}{a+1}$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $1 = \left \frac{a - b}{1 + ab} \right $ $a - b = 1 + ab$ $-b - ab = 1 - a$ $-b(1 + a) = 1 - a$ $b(1 + a) = a - 1$ $b = \frac{a-1}{a+1} \text{ as required}$	} 1 1	

Question	Solutions	Marks	Marking Criteria
2b)	$\sum_{a=1}^n \frac{n-a}{n}$ $= \frac{n-1}{n} + \frac{n-2}{n} + \dots + \frac{n-(n-1)}{n} + \frac{n-n}{n}$ $= 1 - \frac{1}{n} + 1 - \frac{2}{n} + \dots + 1 - \frac{n}{n}$ $= \underbrace{1+1+1+\dots}_{n \text{ times}} - \frac{1}{n} \underbrace{(1+2+3+\dots+n)}_{\substack{\text{AP with } a=1 \\ d=1}}$ $= n + \left(-\frac{1}{n}\right) \left\{ \frac{n}{2} (2a + (n-1)d) \right\}$ $= n + \frac{-1}{n} \left\{ \frac{n}{2} (2 + (n-1)1) \right\}$ $= n + \frac{-1}{n} \left\{ \frac{n}{2} (n+1) \right\}$ $= n - \frac{1}{n} \left\{ \frac{n^2}{2} + \frac{n}{2} \right\}$ $= n - \frac{n}{2} - \frac{1}{2}$ $= \frac{n}{2} - \frac{1}{2}$	1	
2c)	A(1, -3) B(10, 9) P(4, 1) $\therefore x_1 = 1$ $x_2 = 10$ $y_1 = -3$ $y_2 = 9$ $P(4, 1) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$ $\therefore 4 = \frac{m(10) + n(1)}{m+n}; 1 = \frac{m(9) + n(-3)}{m+n}$ $4m + 4n = 10m + n \quad \therefore m+n = 9m-3n$ $3n = 6m \quad \therefore 4n = 8m$ $\therefore \boxed{m = \frac{1}{2}n}$ $\therefore \boxed{m = \frac{1}{2}n}$ $\therefore \frac{m}{n} = \frac{1}{2}$ $\therefore m:n = \frac{1}{2}n : n = 1:2$	1	

Question	Solutions	Marks	Marking Criteria
2d)i)	$x^2 + y^2 - 2x - 14y + 25 = 0$ $y = mx$ intersect when $x^2 + (mx)^2 - 2x - 14(mx) + 25 = 0$ $x^2 + m^2x^2 - 2x - 14mx + 25 = 0$ $(m^2 + 1)x^2 - (2 + 14m)x + 25 = 0$ $(m^2 + 1)x^2 - 2(1 + 7m)x + 25 = 0$ $\therefore \Delta = [-2(1 + 7m)]^2 - 4(m^2 + 1)25$ $= 4(1 + 7m)^2 - 100(m^2 + 1)$ $= 4[(1 + 7m)^2 - 25(m^2 + 1)]$ for 2 real roots, $\Delta > 0$ $\therefore 4[(1 + 7m)^2 - 25(m^2 + 1)] > 0$ $\therefore (1 + 7m)^2 - 25(m^2 + 1) > 0$ as required	1	
ii)	for the line to be a tangent to the curve, $\Delta = 0$ $\therefore (1 + 7m)^2 - 25(m^2 + 1) = 0$ $\therefore 49m^2 + 14m + 1 - 25m^2 - 25 = 0$ $24m^2 + 14m - 24 = 0$ $\therefore 12m^2 + 7m - 12 = 0$ $(3m + 4)(4m - 3) = 0$ $\therefore m = -\frac{4}{3}, \frac{3}{4}$	1	

Question	Solutions	Marks	Marking Criteria
3a)	 when $x = -1, y = 3$ when $x = 2, y = 3$ $\text{Area} = \frac{1}{2}(1\frac{1}{2})(3) + \frac{1}{2}(1\frac{1}{2})(3)$ $= \frac{1}{2} \cdot \frac{9}{2} + \frac{1}{2} \cdot \frac{9}{2}$ $= 2 \cdot \frac{9}{4}$ $= 4\frac{1}{2}$	1	
3b)	$x + \ln x < 2$ $\therefore x + \ln x - 2 = 0$ $x_0 = 1.5$ $f(x) = x + \ln x - 2$ $f'(x) = 1 + \frac{1}{x}$ $f'(1.5) = 1 + \frac{1}{1.5}$ $= 1 + \frac{2}{3}$ $= \frac{5}{3}$ Newton's method: $x_1 = x_0 + \frac{f(x_0)}{f'(x_0)}$ $\therefore x_1 = 1.5 + \frac{1.5 + \ln 1.5 - 2}{\frac{5}{3}}$ $x_1 = 1.5 + \frac{3}{5}(\ln 1.5 - 0.5)$ $x_1 \approx 1.443279065$ $x_1 = 1.44$ (to 2 dec. pl.)	1	

Question	Solutions	Marks	Marking Criteria
3c) i)	$3x^3 + 3x^2 - x - 1$ $= 3x^2(x+1) - (x+1)$ $= (3x^2 - 1)(x+1)$	1	
ii)	$3\tan^3\theta + 3\tan^2\theta - \tan\theta - 1 = 0$ $0^\circ \leq \theta \leq 180^\circ$ $\therefore (3\tan^2\theta - 1)(\tan\theta + 1) = 0$ $\tan^2\theta = \frac{1}{3} \text{ or } \tan\theta = -1$ $\tan\theta = \pm \frac{1}{\sqrt{3}} \quad \theta = 135^\circ$ $\theta = 30^\circ, 150^\circ$ $\therefore \theta = 30^\circ, 135^\circ, 150^\circ$	1	
3d)	$x^4 + 4y^4 = (x^2 + 2y^2 + axy)(x^2 + 2y^2 - axy)$ $\text{RHS: } (x^2 + 2y^2 + axy)(x^2 + 2y^2 - axy)$ $= [(x^2 + 2y^2)^2 - (axy)^2]$ $= x^4 + 4x^2y^2 + 4y^4 - a^2x^2y^2$ $= x^4 + 4y^4 + 4x^2y^2 - a^2x^2y^2$ $= x^4 + 4y^4 + (4 - a^2)x^2y^2$ $= \text{LHS}$ $= x^4 + 4y^4$ $\therefore 4 - a^2 = 0$ $\therefore (2 - a)(2 + a) = 0$ $\therefore a = \pm 2$	1	

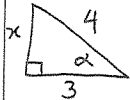
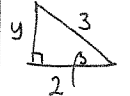
Question	Solutions	Marks	Marking Criteria
4a)	$y = 2^x \quad P(1, 2)$ $\frac{dy}{dx} = \ln 2 \cdot 2^x$ $\text{at } x = 1, \frac{dy}{dx} = \ln 2 \cdot 2^1 = 2 \ln 2$ $\text{eqn of tangent: } y - 2 = 2 \ln 2 (x - 1)$ $y - 2 = (2 \ln 2)x - 2 \ln 2$ $y - 2 = (\ln 2^2)x - \ln 2^2$ $y - 2 = (\ln 4)x - \ln 4$ $y = x \ln 4 - \ln 4 + 2$	1	
4b)	$y = \frac{1}{x}$  $\therefore \int_k^e \frac{1}{x} dx = 1$ $[\ln x]_k^e = 1$ $\ln e - \ln k = 1$ $1 - \ln k = 1$ $\therefore -\ln k = 0$ $\therefore \ln k = 0$ $\therefore k = 1 \quad (\text{since } e^0 = 1)$	1	

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Question	Solutions	Marks	Marking Criteria
4c) i)	$1 \times 1 + (\sqrt{3}-1)(\sqrt{3}+1) + (\sqrt{3}-1)^2(\sqrt{3}+1)^2 \dots$ $= 1 + 2 + 4 + \dots$ $\frac{T_2}{T_1} = \frac{2}{1}$ $\frac{T_3}{T_2} = \frac{4}{2}$ $\therefore \frac{T_3}{T_2} = \frac{T_2}{T_1}$ $\therefore$ the series is geometric with $a=1, r=2$	1	
ii)	$T_n = ar^{n-1}$ $= (1)(2)^{n-1}$ $\therefore T_n = 2^{n-1}$	1	
4d)	Step 1: test for $n=1$ $13 \times 6^1 + 2$ $= 78 + 2$ $= 80$, which is divisible by 5 $\therefore$ true for $n=1$ Step 2: assume true for all $n=k$ $\therefore 13 \times 6^k + 2 = 5m$ for some integer m Step 3: prove true for $n=k+1$ $13 \times 6^{k+1} + 2$ $= 13 \times 6^k \cdot 6 + 2$ $= 6[5m - 2] + 2$ $= 30m - 12 + 2$ $= 30m - 10$	1	

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Question	Solutions	Marks	Marking Criteria
	$= 5(6m-2)$, which is divisible by 5 $\therefore$ true for $n=k+1$	1	
	Step 4: The statement is true for $n=k+1$ whenever it is true for $n=k$ But the statement is true for $n=1$ $\therefore$ By the Principle of Mathematical Induction the statement is true for all integers $n \geq 1$	1	
5a)	$2x^3 - 5x^2 + 7x - 1 = 0$		
(i)	$\alpha + \beta + \gamma = \frac{5}{2}$	1	
(ii)	$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{7}{2}$	1	
(iii)	$\alpha\beta\gamma = \frac{1}{2}$ $(\alpha-1)(\beta-1)(\gamma-1)$ $= (\alpha\beta - \alpha - \beta + 1)(\gamma-1)$ $= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$ $= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1$ $= \frac{1}{2} - \frac{7}{2} + \frac{5}{2} - 1$ $= -\frac{3}{2}$	1	

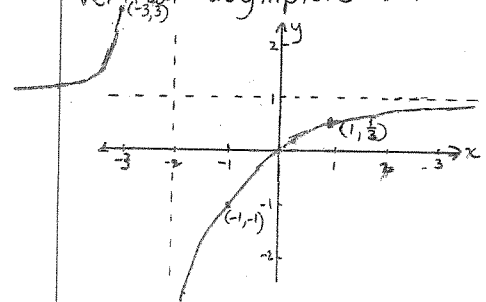
Question	Solutions	Marks	Marking Criteria
5b)	$\cos \alpha = \frac{3}{4}, 0^\circ < \alpha < 90^\circ$ $\cos \beta = \frac{2}{3}, 270^\circ < \beta < 360^\circ$  $x^2 + 3^2 = 4^2$ (by Pythagoras' Theorem) $x^2 + 9 = 16$ $x = \sqrt{7}$ $\therefore \sin \alpha = \frac{\sqrt{7}}{4}$  $y^2 + 2^2 = 3^2$ (by Pythagoras' Theorem) $y^2 + 4 = 9$ $y = \sqrt{5}$ $\therefore \sin \beta = \frac{\sqrt{5}}{3}$ but β lies in Quadrant 4 $\therefore \sin \beta = -\frac{\sqrt{5}}{3}$ $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ $= \left(\frac{\sqrt{7}}{4}\right)\left(\frac{2}{3}\right) - \left(\frac{3}{4}\right)\left(-\frac{\sqrt{5}}{3}\right)$ $= \frac{2\sqrt{7}}{12} + \frac{3\sqrt{5}}{12}$ $= \frac{\sqrt{7}}{6} + \frac{\sqrt{5}}{4} \quad (\text{or } \frac{2\sqrt{7} + 3\sqrt{5}}{12})$		

Question	Solutions	Marks	Marking Criteria
5c)	$P(2ap, ap^2)$ $x^2 = 4ay$ i) normal: $x + py = 2ap + ap^3$ Q lies on y -axis, $\therefore x = 0$ $\therefore 0 + py = 2ap + ap^3$ $py = ap(2 + p^2)$ $\therefore y = a(2 + p^2), p \neq 0$ $\therefore Q$ is the point $(0, a(2 + p^2))$ ii) Q is the midpoint of PR $\therefore (0, a(2 + p^2)) = \left(\frac{2ap + x_2}{2}, \frac{ap^2 + y_2}{2}\right)$ $\therefore \frac{2ap + x_2}{2} = 0$ $\therefore 2ap + x_2 = 0$ $\therefore \boxed{x_2 = -2ap}$ $\frac{ap^2 + y_2}{2} = a(2 + p^2)$ $\therefore ap^2 + y_2 = 2a(2 + p^2)$ $ap^2 + y_2 = 4a + 2ap^2$ $\therefore \boxed{y_2 = 4a + ap^2}$ $\therefore R$ is the point $(-2ap, ap^2 + 4a)$ iii) $x = -2ap, y = ap^2 + 4a$ $\therefore p = \frac{x}{-2a} \therefore p = \sqrt{\frac{y - 4a}{a}}$ $\therefore \frac{x^2}{4a^2} = \frac{y - 4a}{a}$		

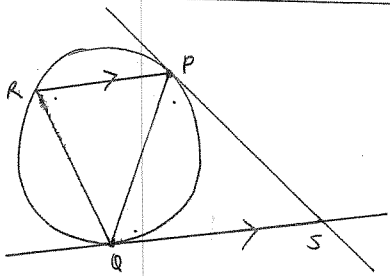
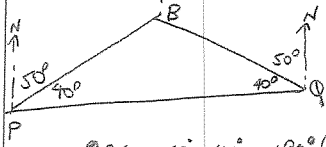
Qn	Solutions	Marks	Comments+Criteria
	$d_{ap}^2 = (2ap - 0)^2 + (ap^2 - 2a - ap^2)^2$ $= 4a^2p^2 + a^2p^4 - 2a^2p^2 - a^2p^4$ $- 2a^3p^2 + 4a^2 + 2a^2p^2$ $= 2a^2p^2 + 4a^2$		

Qn	Solutions	Marks	Comments+Criteria
	$R(-2ap, ap^2 + 4a)$ $Q(0, a(2 + p^2))$ $P(2ap, ap^2)$ then $d_{ae} = d_{ra}$. $d_{ae}^2 = (-2ap - 0)^2 + (ap^2 + 4a - 2a - ap^2)^2$ $= 4a^2p^2 + (ap^2 + 2a - ap^2)^2$ $= 4a^2p^2 + a^2(p^2 + 2a - ap^2)^2$ $= 4a^2p^2 + a^2(p^4 + 2ap^2 - ap^4 + 2ap^2 + 4a^2$ $- 2a^2p^2 - ap^4 - 2a^2p^2 - a^2p^4)$ $= 4a^2p^2 + a^2(p^4 + 4ap^2 - 2ap^4 + 4a^2$ $- 4a^2p^2 - a^2p^4)$ $= 4a^2p^2 + a^2p^4 + 4a^3p^2 - 2a^3p^4 + 4a^4$ $- 4a^4p^2 - a^4p^2$		

Question	Solutions	Marks	Marking Criteria
5c)iii)cd)	$x^2 = \frac{4a^2(y-4a)}{a}$ $x^2 = 4a(y-4a)$ $\therefore$ locus is a parabola with focal length a and vertex $(0, 4a)$	1	
6a)	$\frac{d}{dx} \frac{4x+1}{2x-3}$ $= \frac{(2x-3)(4) - (4x+1)(2)}{(2x-3)^2}$ $= \frac{8x-12-8x-2}{(2x-3)^2}$ $= \frac{-14}{(2x-3)^2}$ $\therefore \int_0^1 \frac{dx}{(2x-3)^2}$ $= -\frac{1}{14} \int \frac{-14}{(2x-3)^2} dx$ $= -\frac{1}{14} \left[\frac{4x+1}{2x-3} \right]_0^1$ $= -\frac{1}{14} \left\{ \left(\frac{4+1}{2-3} \right) - \left(\frac{0+1}{0-3} \right) \right\}$ $= -\frac{1}{14} \left\{ \frac{5}{-1} - \frac{1}{-3} \right\}$ $= -\frac{1}{14} \left(-5 + \frac{1}{3} \right)$ $= -\frac{1}{14} \left(-\frac{14}{3} \right)$ $= -\frac{1}{14} \cdot -\frac{14}{3}$ $= \frac{1}{3}$	1	

Question	Solutions	Marks	Marking Criteria
6b)	$f(x) = \frac{x}{x+2}$		
i)	$f'(x) = \frac{(x+2)(1) - x(1)}{(x+2)^2}$ $= \frac{x+2-x}{(x+2)^2}$ $= \frac{2}{(x+2)^2} > 0 \text{ for all } x$ since $(x+2)^2 > 0$ for all x	1	
ii)	$f(x) = \frac{x}{x+2}$ $= \frac{x+2-2}{x+2}$ $= 1 - \frac{2}{x+2}$ $\therefore$ horizontal asymptote at $y=1$ (since $\frac{2}{x+2} \neq 0$ for all x)	1	
iii)	$f'(x) > 0 \text{ for all } x$ $\therefore f(x)$ is increasing for all x horizontal asymptote at $y=1$ Vertical asymptote at $x=-2$  When $x=0$, $y=0$ When $x=-3$, $y=3$	1	1 for graph 1 for some working

Question	Solutions	Marks	Marking Criteria
6 iv)		1	
v)	$x = \frac{y}{y+2}$ $\therefore xy + 2x = y$ $\therefore xy - y = -2x$ $y(x-1) = -2x$ $y = \frac{-2x}{x-1}$ $\therefore y = \frac{2x}{1-x}$ $\therefore f^{-1}(x) = \frac{2x}{1-x}$	1	
vi)	Domain $f^{-1}(x) = \{x: \text{all real } x, x \neq 1\}$	1	
7a) i)	$\sin A = \frac{2t}{1+t^2}, \cos A = \frac{1-t^2}{1+t^2}$ $\text{let } t = \tan A$	1	
ii)	$\text{RTP: } \frac{\sin 2A}{1 + \cos 2A} = \tan A$ $\text{LHS: } \frac{2t}{1+t^2}$ $1 + \frac{1-t^2}{1+t^2}$ $= \frac{2t}{1+t^2}$ $\frac{1+t^2+1-t^2}{1+t^2}$ $= \frac{2t}{2}, 1+t^2 \neq 0$ $= t = \tan A = \text{RHS}$	1	

Question	Solutions	Marks	Marking Criteria
7b)	 $RTP: QP = QR$ $PS = QS$ (equal tangents from external point S) $\therefore \angle SPQ = \angle SQP$ (base angles of isosceles $\triangle PQS$ equal) $\angle PQS = \angle QRP$ (angle in alternate segment is equal) $\angle PQS = \angle QPR$ (alternate angles equal; $PR \parallel QS$) $\therefore \angle QRP = \angle QPR$ $\therefore \triangle QPR$ is isosceles (base angles equal) $\therefore QP = QR$ as required		
7c) i)	 $\therefore \angle PBQ + 40^\circ + 40^\circ = 180^\circ$ (angle sum $\triangle PQR$) $\therefore \angle PBQ = 100^\circ$ as required		

Question	Solutions	Marks	Marking Criteria
7c) ii)	$\tan 30 = \frac{h}{PB}$ $\tan 45 = \frac{h}{BQ}$ $PB = \frac{h}{\tan 30}$ $BQ = \frac{h}{\tan 45}$ $PB = h \cot 30$ $BQ = h \cot 45$	1	
iii)	$1000^2 = h^2 \cot^2 30 + h^2 \cot^2 45 - 2h^2 \cot 30 \cot 45 \cos 100^\circ$ $\therefore h^2 = \frac{1000^2}{\cot^2 30 + \cot^2 45 - 2 \cot 30 \cot 45 \cos 100^\circ}$		
iv)	$h^2 = \frac{1000^2}{3 + 1 - (2\sqrt{3})(1)(-0.173648177)}$ $h^2 = \frac{1000^2}{3.398465067}$ $h \div 17.15373098$ $h = 17 \text{ m (to nearest metre)}$ $PQ^2 = PB^2 + BQ^2 - 2PB \cdot BQ \cdot \cos \angle PBQ$ $1000^2 = h^2 \cot^2 30 + h^2 \cot^2 45 - 2h \cot 30 \cdot h \cot 45 \cos 100^\circ$		